

PROJECT REPORT

# VerifyX AI

AI-Based Fake Identity & Document Screening System

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Submitted by

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*Verify the Document. Verify the Identity*

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# 1. Introduction and Problem Definition

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## 1.1 Abstract

VerifyX AI is an AI-assisted identity document screening system designed to analyse identity documents and provide a structured verification assessment. The system accepts documents such as Aadhaar and PAN, identifies the document type, extracts information using Optical Character Recognition (OCR), performs the document and identity checks that are available, and generates a risk assessment.

The system also supports live-photo and liveness verification where applicable. Instead of relying on a single verification result, VerifyX AI combines multiple available signals and identifies whether a result can be accepted automatically or requires further review by a human reviewer.

The platform is built with a Django backend and an HTML, CSS and JavaScript frontend, and uses OCR, OpenCV-based image processing and machine-learning components for analysis. Results are stored in a SQLite or PostgreSQL database so that every screening can be revisited through a history module with date-range filtering.

A central design principle is that VerifyX AI is a screening and decision-support tool. A risk indication does not by itself prove that a document is fraudulent, and a missing check (for example, an unreadable QR code) is reported as unavailable rather than as evidence of fraud. Final official verification remains with the appropriate authorised authority or service.

### 1.1.1 Keywords

Identity verification, document screening, OCR, Aadhaar, PAN, liveness detection, face verification, risk assessment, explainable AI, Django, OpenCV.

## 1.2 Introduction

### 1.2.1 Background

Identity documents are commonly used for verification in banks, educational institutions, companies, government services and many other organisations. Whenever a person opens an account, joins an institution, applies for a service or completes a know-your-customer process, someone must confirm that the document is genuine and that it belongs to the person presenting it.

Manual verification of these documents takes considerable time and depends heavily on the experience and attention of the reviewer. It becomes harder still when documents are photographed in poor light, scanned at low resolution, partially obscured, or when they contain subtle modifications that are difficult to see with the naked eye.

### 1.2.2 Motivation

As the volume of digital submissions grows, organisations need a way to perform a fast and consistent first-level check so that reviewers can concentrate their time on the documents that truly need

attention. A screening system that is transparent about what it checked, what it could not check, and why it reached a result can make this preliminary step both quicker and more reliable.

### 1.2.3 Overview of VerifyX AI

VerifyX AI provides a digital screening approach in which an uploaded document is analysed using OCR, image processing, machine-learning components and verification checks. The system produces a structured result in one of the following forms:

- Verified
- Review Required
- Mismatch / Inconsistent
- Risk Assessment (an overall score and status)

The purpose of the system is to assist document screening, not to replace authorised government or institutional verification.

### 1.2.4 Organisation of this Report

This report is organised into seven chapters. Chapter 1 covers the abstract, introduction, problem statement and objectives. Chapter 2 covers scope, key concepts, the proposed solution and requirements. Chapter 3 covers architecture, modules, workflow and the AI implementation. Chapter 4 covers RAG, risk assessment, database design, the user interface and history management. Chapter 5 covers security, testing and use cases. Chapter 6 covers implementation, data flow, deployment, illustrative scenarios, advantages, limitations and challenges. Chapter 7 covers future scope, the conclusion and appendices.

## 1.3 Problem Statement

Traditional identity document checking faces several practical challenges. Each challenge below affects speed, accuracy or consistency, and together they show why a structured, evidence-based screening layer is valuable.

| Challenge                                                 | Impact                                               | How VerifyX AI responds                                  |
|-----------------------------------------------------------|------------------------------------------------------|----------------------------------------------------------|
| Manual verification requires time                         | Queues build up and turnaround slows as volumes grow | Automates the preliminary screening steps                |
| Different document types have different structures        | Reviewers must remember many layouts and rules       | Detects document type and applies type-specific analysis |
| Text is difficult to read from images or scans            | Data entry errors and missed details                 | Uses OCR to extract text automatically                   |
| Modified or suspicious documents need detailed inspection | Subtle changes can be overlooked                     | Analyses document characteristics for inconsistent areas |
| A genuine document may be presented by a different person | A valid document alone does not prove identity       | Adds liveness and face verification where applicable     |

| Challenge                                                 | Impact                             | How VerifyX AI responds                              |
|-----------------------------------------------------------|------------------------------------|------------------------------------------------------|
| Results from different checks must be considered together | Isolated results may be misleading | Combines signals into a single structured assessment |

### 1.3.1 Need for a Preliminary Screening System

There is a need for a system that can perform preliminary automated screening and present the evidence clearly to the reviewer. Such a system should not simply return a yes or no answer. It should show which checks passed, which checks raised concerns, and which checks could not be performed, so that the reviewer can make an informed decision quickly.

### 1.3.2 Problem Definition

Given an uploaded identity document, and optionally a live photograph of the person presenting it, the system must (a) identify the document type, (b) extract its textual content, (c) evaluate its integrity and internal consistency, (d) evaluate whether the presenter matches the document photograph, and (e) combine all available evidence into a clear status and risk score, while recording the outcome for later review.

## 1.4 Objectives

The main objectives of VerifyX AI are listed below.

1. To provide a digital platform for identity document screening.
2. To identify the type of uploaded document.
3. To extract document information using OCR.
4. To analyse document characteristics for suspicious changes.
5. To perform available QR and document consistency checks.
6. To perform liveness and identity checks where applicable.
7. To generate a structured risk assessment.
8. To provide explanations for results requiring further review.
9. To maintain verification history.
10. To support future expansion to additional document types and countries.

### 1.4.1 Expected Outcomes

- **Reduced effort:** Reviewers spend less time on documents that pass all checks clearly.
- **Better prioritisation:** Documents with higher risk indications reach reviewers first.
- **Transparency:** Every result is accompanied by the signals that produced it.
- **Traceability:** Each screening is stored with its result, score and status.
- **Extensibility:** New document types and rules can be added without redesigning the system.

## 2. Scope, Concepts, Solution and Requirements

### 2.1 Scope of the Project

The current system focuses on identity document screening, particularly documents such as Aadhaar, PAN and other supported identity documents. The system can be extended to support additional documents such as passports and driving licences by adding document-specific rules and verification methods.

#### 2.1.1 In Scope and Out of Scope

| In scope                                                              | Out of scope                                              |
|-----------------------------------------------------------------------|-----------------------------------------------------------|
| Upload of PDF and image documents through a web interface             | Issuing or replacing identity documents                   |
| Document type detection and OCR extraction                            | Final official verification with issuing authorities      |
| Document integrity analysis and QR consistency checks where available | Declaring a document fake on the basis of a single signal |
| Liveness and face verification where applicable                       | Guaranteeing authenticity of every document               |
| Risk assessment, explanations and verification history                | Replacing authorised institutional verification services  |

#### 2.1.2 Assumptions

- Users upload documents that they are entitled to submit for verification.
- Where liveness and face checks are used, the user grants camera access and a live capture is possible.
- Reviewers use the results as decision support and remain responsible for final decisions.

#### 2.1.3 Intended Users

The intended users are reviewers and administrators in organisations such as banks, educational institutions, companies and service providers who need to screen identity documents before a final decision is made.

The system is intended as a screening and decision-support tool. Final official verification should be performed through the appropriate authorised authority or service.

## 2.2 Background and Key Concepts

This chapter introduces the main technical ideas on which VerifyX AI is built. Each concept is described at a level that explains its role in the system.

### 2.2.1 Optical Character Recognition (OCR)

OCR stands for Optical Character Recognition. It converts text that appears in an image or scanned page into machine-readable text. In VerifyX AI, OCR reads fields such as name, date of birth, document number and address so that they can be analysed without manual data entry. OCR accuracy depends

on image quality, so the system treats low-confidence extraction as a reason for review rather than as proof of a problem.

### **2.2.2 Document Classification**

Document classification decides what kind of document has been uploaded, for example Aadhaar, PAN or another supported type. Knowing the type lets the system apply the correct field layout, expected formats and verification rules.

### **2.2.3 Image Processing with OpenCV**

OpenCV is a widely used library for image processing. Typical operations include resizing, denoising, contrast adjustment, rotation correction and region detection. These operations improve the quality of the input to OCR and support the analysis of visual characteristics of the document.

### **2.2.4 Document Integrity Analysis**

Document integrity analysis examines available characteristics of the document to find areas that appear inconsistent or suspicious. Its output is a risk indication such as Low Risk, Medium Risk or Review Required. It is important to note that a risk indication does not by itself prove that a document is fraudulent; it is a signal that helps decide where a reviewer should look.

### **2.2.5 QR Code Analysis**

Some identity documents carry a QR code that encodes information about the document. When a supported QR code is present and readable, its content can be compared with the information extracted by OCR. If the QR code cannot be processed, the system reports the result as Unavailable / Not Available instead of treating the missing result as evidence of fraud.

### **2.2.6 Liveness Verification**

Liveness verification checks whether a live capture comes from a real, present person rather than from a static image or a replayed photograph. In VerifyX AI, a live camera capture is analysed and the outcome is reported as Passed or Review Required.

### **2.2.7 Face Verification**

Face verification compares the face in the document photograph with the face in the live photograph. The possible results are Verified, Review Required and Mismatch. Review Required means that the system does not have sufficient confidence for an automatic decision; it does not automatically mean that the document is fake.

### **2.2.8 Risk Assessment and Explainability**

Risk assessment combines the available signals into an overall score and status. Explainability means that the system shows the main signals behind the result, so that a reviewer can understand and, where necessary, challenge it. The risk score exists to help prioritise review, not to make an unsupported final determination.

## **2.3 Proposed Solution**

VerifyX AI provides a centralised system for identity document screening. A document is uploaded, its type is detected, its content is extracted and analysed, available liveness and identity checks are

performed, and a risk assessment produces the final verification result, which is then saved in the history.

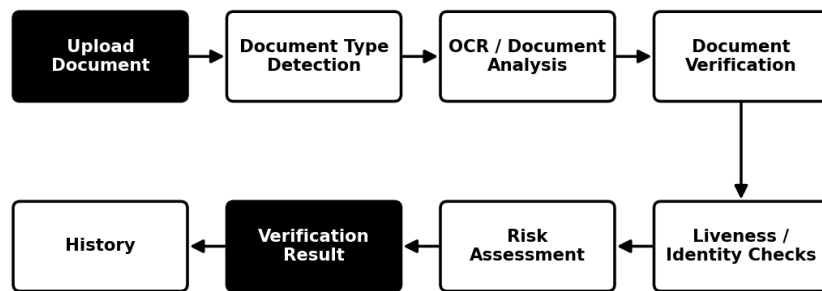


Figure 2.1: Basic processing pipeline of VerifyX AI

### 2.3.1 Design Principles

- **Evidence-based:** Results are built from several signals rather than from one check.
- **Non-accusatory:** The system reports risk and inconsistency, not accusations of fraud.
- **Honest about gaps:** Checks that cannot be performed are shown as unavailable.
- **Human in the loop:** Uncertain results are routed to a reviewer.
- **Extensible:** Document-specific rules can be added for new document types.

### 2.3.2 Verification Outcomes

| Outcome                 | Meaning                                                       | Typical action                                  |
|-------------------------|---------------------------------------------------------------|-------------------------------------------------|
| Verified                | Available signals are consistent and confidence is sufficient | Can be accepted automatically                   |
| Review Required         | Some signals lack confidence or need inspection               | Send to a reviewer with the supporting evidence |
| Mismatch / Inconsistent | Signals conflict with one another                             | Reviewer investigates before any decision       |

The system analyses the available information and presents the result in a simple verification dashboard so that the reviewer can see at a glance which part of the verification requires attention.

## 2.4 System Requirements

### 2.4.1 Functional Requirements

| ID  | Requirement                                                                                                |
|-----|------------------------------------------------------------------------------------------------------------|
| FR1 | The system shall allow users to upload PDF, JPG, JPEG, PNG and WEBP documents.                             |
| FR2 | The system shall enforce a configurable maximum upload size.                                               |
| FR3 | The system shall detect the type of the uploaded document.                                                 |
| FR4 | The system shall extract name, date of birth, document number, address and other available text using OCR. |

| ID   | Requirement                                                                                                   |
|------|---------------------------------------------------------------------------------------------------------------|
| FR5  | The system shall analyse document integrity and report Low Risk, Medium Risk or Review Required.              |
| FR6  | The system shall perform QR analysis where a supported QR code is available and report Unavailable otherwise. |
| FR7  | The system shall perform liveness verification from a live camera capture where applicable.                   |
| FR8  | The system shall compare document and live photographs where both are available.                              |
| FR9  | The system shall generate a risk score, a status and an explanation.                                          |
| FR10 | The system shall store screening records and provide a history with date-range filtering.                     |

### 2.4.2 Non-Functional Requirements

| Quality         | Requirement                                                       |
|-----------------|-------------------------------------------------------------------|
| Usability       | A clear, simple interface for upload, results and history         |
| Performance     | Results returned within a time acceptable for interactive review  |
| Reliability     | Graceful handling of unreadable, blurred or rotated documents     |
| Security        | Protected transport and storage of documents and personal data    |
| Maintainability | Modular services so that components can be improved independently |
| Extensibility   | Ability to add document types, rules and countries                |
| Transparency    | Every result accompanied by the signals behind it                 |

### 2.4.3 Software Requirements

| Layer                  | Technology                                                         |
|------------------------|--------------------------------------------------------------------|
| Frontend               | HTML, CSS, JavaScript                                              |
| Backend                | Python, Django                                                     |
| AI / ML and processing | Machine learning, OCR, OpenCV, image processing, document analysis |
| Database               | SQLite (development) or PostgreSQL (deployment)                    |
| Client                 | A modern web browser with camera access for live capture           |

### 2.4.4 Hardware and Network Requirements

For development and demonstration, a standard laptop or desktop computer is sufficient. For deployment, the server should have enough memory and processing capacity for image processing and OCR, and a reliable network connection is needed so that documents and results move quickly between the browser and the server. Image and model processing can benefit from additional CPU capacity, and deployments with high volumes can add separate processing workers as described in the deployment chapter.

## 3. Architecture, Modules, Workflow and AI

### 3.1 System Architecture

VerifyX AI follows a layered architecture in which a lightweight web frontend communicates with a Django backend. The backend coordinates OCR, document analysis and verification services, passes their outputs to the risk assessment component, and stores results in the database.

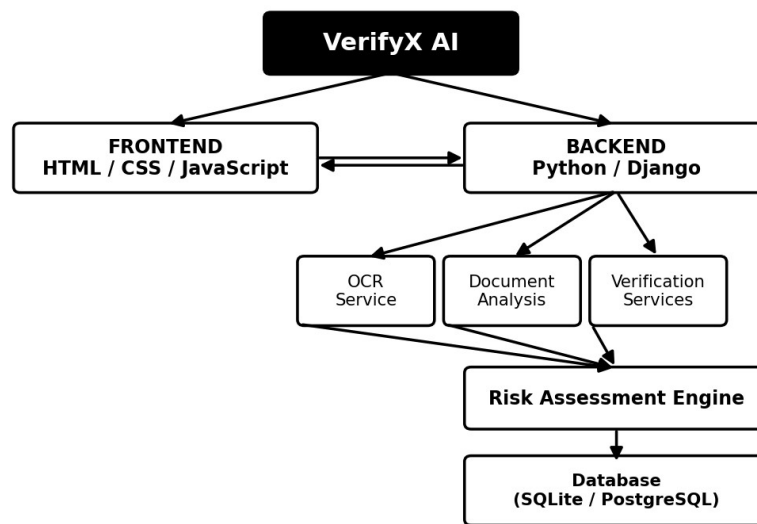


Figure 3.1: High-level architecture of VerifyX AI

#### 3.1.1 Frontend Layer

The frontend, built with HTML, CSS and JavaScript, provides the user interface for document upload, live capture, verification results and history. JavaScript handles interaction such as file selection, camera access and the display of results.

#### 3.1.2 Backend Layer

Django manages application logic, APIs, document processing requests, verification workflows and database communication. Keeping the workflow in the backend means that all processing rules are applied consistently regardless of how the frontend is presented.

#### 3.1.3 Processing Services

- **OCR service:** Converts document images into machine-readable text.
- **Document analysis:** Classifies the document and examines its visual characteristics.
- **Verification services:** Perform QR, liveness, face and identity consistency checks.
- **Risk assessment:** Combines all available signals into a score and status.

### **3.1.4 Data Layer**

The database stores application data and verification history. SQLite is convenient for development and demonstration, while PostgreSQL is suitable for multi-user deployment.

### **3.1.5 Component Interaction**

A typical request begins when the frontend sends an uploaded file to the backend. The backend validates the file, requests document classification, runs OCR and document analysis, invokes the applicable verification services, and forwards the collected evidence to the risk assessment component. The resulting status, score and explanation are saved and returned to the frontend for display.

## **3.2 Module Description**

This chapter describes each main feature of VerifyX AI in detail.

### **3.2.1 Document Upload**

Users upload supported identity documents through the web interface. The system can be configured to support PDF, JPG, JPEG, PNG and WEBP files, and a maximum upload size can be enforced to prevent unnecessarily large files. Basic validation of file type and size happens before any analysis begins.

### **3.2.2 Document Type Detection**

The system identifies the type of the uploaded document and applies the relevant analysis process. The uploaded document is classified as Aadhaar, PAN or another supported type, and this decision selects the field layout and rules used in the subsequent steps.

### **3.2.3 OCR**

OCR extracts readable information from the uploaded document, such as name, date of birth, document number, address and other available text. It allows the system to process information from document images instead of requiring users to enter the information manually.

### **3.2.4 Document Integrity Analysis**

The system analyses available document characteristics to identify suspicious or inconsistent areas. The result can be represented as Low Risk, Medium Risk or Review Required. A risk indication does not by itself prove that a document is fraudulent.

### **3.2.5 QR Verification**

For documents containing a supported QR code, the system can perform QR-related analysis when the required information is available. If the QR cannot be processed, the result shows Unavailable / Not Available instead of treating the missing QR result as proof of fraud.

### 3.2.6 Liveness Verification

The system can use a live camera capture to determine whether the captured image passes the available liveness check. The outcome is Passed or Review Required.

### 3.2.7 Face Verification

Where both a document photograph and a live photograph are available, the system compares the available facial information. Possible results are Verified, Review Required and Mismatch. Review Required means the system does not have sufficient confidence for an automatic decision and does not automatically mean the document is fake.

### 3.2.8 Identity Consistency

The system considers the available verification signals and checks whether they are consistent with one another. Document information, OCR information and the available identity checks are evaluated together to produce an identity consistency outcome.

### 3.2.9 Risk Assessment

The system combines the available verification signals and generates an overall assessment, for example a risk score of 64 out of 100 with the status Manual Review and the reason that some verification signals require further inspection. The purpose of the risk score is to help prioritise review rather than to make an unsupported final determination.

### 3.2.10 Verification History

The system stores previous screening records. The History page can provide document ID, document type, date, result, risk score and verification status, and a date-range filter can be applied.

### 3.2.11 Module Summary

| Module               | Input                    | Output                                   |
|----------------------|--------------------------|------------------------------------------|
| Upload               | PDF or image file        | Validated document                       |
| Type detection       | Validated document       | Aadhaar / PAN / Other                    |
| OCR                  | Document image           | Name, DOB, number, address, other text   |
| Integrity analysis   | Document characteristics | Low Risk / Medium Risk / Review Required |
| QR verification      | QR code (if present)     | Result or Unavailable                    |
| Liveness             | Live capture             | Passed / Review Required                 |
| Face verification    | Document and live photos | Verified / Review Required / Mismatch    |
| Identity consistency | All available signals    | Consistent / Inconsistent                |
| Risk assessment      | All available signals    | Score, status, explanation               |
| History              | Saved screening records  | Filtered list and counts                 |

### 3.3 Verification Workflow

The verification workflow describes how a single document moves through the system from upload to saved result.

1. Start: the user uploads a document.
2. Identify document type.
3. Extract text with OCR.
4. Perform document analysis.
5. Run the QR check, liveness check and identity check in parallel where they apply.
6. Combine the outputs in evidence analysis.
7. Produce the risk assessment.
8. Assign an outcome: Verified, Review Required or Mismatch.
9. Save the record to the history.

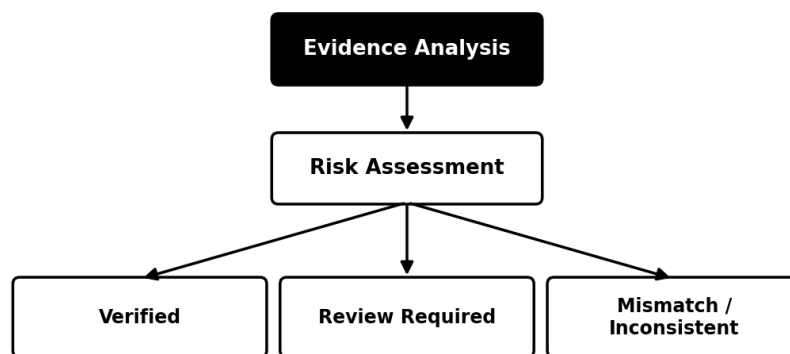


Figure 3.2: Evidence analysis leading to one of three outcomes

#### 3.3.1 Handling Unavailable Checks

Not every check is possible for every document. A QR code may be missing or unreadable, or a live photograph may not be supplied. In these situations the corresponding check is marked as unavailable and the remaining signals are used. The absence of a signal is recorded in the explanation so the reviewer knows exactly what was and was not evaluated.

#### 3.3.2 Handling Poor-Quality Input

Blur, poor lighting, rotation and damage can reduce OCR and analysis quality. Rather than returning a confident result from weak evidence, the system lowers confidence and steers the document towards Review Required, ideally with a note recommending that the user upload a clearer image.

### 3.3.3 Routing Decisions

| Situation                                                | Typical routing                                |
|----------------------------------------------------------|------------------------------------------------|
| All available signals consistent, confidence sufficient  | Verified                                       |
| One or more signals lack confidence                      | Review Required                                |
| Signals conflict, for example face or data inconsistency | Mismatch / Inconsistent, reviewer investigates |

## 3.4 AI Implementation

AI and machine learning are used as part of the document screening pipeline. The major AI-related components are described below.

### 3.4.1 Document Analysis

The system analyses the uploaded document and the available visual information. This includes classifying the document type and examining visual characteristics to identify areas that appear inconsistent.

### 3.4.2 Machine Learning

Machine-learning components can be used to identify document patterns and to generate risk-related signals. These signals are inputs to the risk assessment and are never treated as conclusive proof on their own.

### 3.4.3 OCR

OCR converts document images into machine-readable text. Its output feeds consistency checks and the explanation layer.

### 3.4.4 Risk Assessment

Different verification signals are combined to produce a structured risk assessment consisting of a score, a status and a reason.

### 3.4.5 Explainable Results

The system presents the important signals contributing to the result instead of showing only a final score. Explainability builds reviewer trust and makes it easier to detect and correct weaknesses in individual components.

### 3.4.6 Responsible Use of AI

- Predictions are screening evidence, not conclusive proof.
- Uncertainty is surfaced through the Review Required outcome.
- The reviewer, not the model, makes the final decision on uncertain cases.

## 4. RAG, Risk Assessment, Database and Interface

### 4.1 RAG Implementation

This chapter applies if the Retrieval-Augmented Generation (RAG) pipeline has been implemented in the project. RAG stands for Retrieval-Augmented Generation. In VerifyX AI, RAG can provide document-specific verification knowledge to the explanation layer.

#### 4.1.1 RAG Flow

1. Document analysis is completed.
2. The document type is identified.
3. Relevant knowledge is retrieved.
4. Evidence and knowledge are combined.
5. An explanation is generated.

#### 4.1.2 Example

When an Aadhaar document is detected, the system can retrieve relevant reference information associated with Aadhaar and use it to explain the available verification results. For example, it can describe which fields are expected on that document type and which of them were extracted successfully.

#### 4.1.3 Role and Limits of RAG

RAG should support the explanation process rather than independently declaring a document fake. The verification decision is produced by the risk assessment; RAG improves the clarity and context of the explanation given to the reviewer.

| Step                   | Purpose                                                 |
|------------------------|---------------------------------------------------------|
| Identify document type | Select the right knowledge collection                   |
| Retrieve knowledge     | Bring in document-specific reference information        |
| Combine with evidence  | Relate the reference information to the actual findings |
| Generate explanation   | Present a clear, readable reason for the result         |

### 4.2 Risk Assessment Methodology

The risk assessment combines available verification signals into an overall score and status. Its purpose is to help prioritise review rather than to make an unsupported final determination.

#### 4.2.1 Input Signals

| Signal             | Possible values                        |
|--------------------|----------------------------------------|
| Document type      | Aadhaar, PAN, other, not recognised    |
| OCR                | Successful, Review Required            |
| QR                 | Consistent, inconsistent, Unavailable  |
| Document integrity | Low Risk, Medium Risk, Review Required |

| Signal               | Possible values                     |
|----------------------|-------------------------------------|
| Document photo       | Detected, not detected              |
| Liveness             | Passed, Review Required             |
| Face verification    | Verified, Review Required, Mismatch |
| Identity consistency | Consistent, Inconsistent            |

#### 4.2.2 Scoring Approach

Each signal contributes to a score between 0 and 100, where a higher score indicates a greater need for review. Signals that are unavailable are not counted as evidence of fraud; instead they may reduce confidence. The exact weights are a configuration decision and can be tuned as more data becomes available.

#### 4.2.3 Illustrative Status Bands

The bands below are an example of how a score can be mapped to a status. Actual thresholds should be set by the organisation using the system.

| Example score range | Status                  | Meaning                                         |
|---------------------|-------------------------|-------------------------------------------------|
| Lower range         | Verified                | Signals consistent; confidence sufficient       |
| Middle range        | Manual Review           | Some signals require further inspection         |
| Upper range         | Mismatch / Inconsistent | Signals conflict; detailed investigation needed |

#### 4.2.4 Example

A document receives a risk score of 64 out of 100 and the status Manual Review. The reason recorded is that some verification signals require further inspection. The reviewer opens the result, sees which signals raised concerns, and decides on the next step.

#### 4.2.5 Keeping the Score Honest

- The score prioritises review; it does not declare fraud.
- The explanation always lists the signals that contributed.
- Weights and thresholds are reviewed periodically against reviewer decisions.

### 4.3 Database Design

The database stores application and verification history information. SQLite is used for development and PostgreSQL is suitable for deployment. The tables below describe a proposed logical design.

#### 4.3.1 Screening Record

| Field         | Description                               |
|---------------|-------------------------------------------|
| Document ID   | Unique identifier of the screening        |
| Document type | Detected type, for example Aadhaar or PAN |
| Date and time | When the screening was performed          |

| Field                | Description                               |
|----------------------|-------------------------------------------|
| OCR outcome          | Successful or Review Required             |
| QR outcome           | Result or Unavailable                     |
| Integrity result     | Low Risk, Medium Risk or Review Required  |
| Liveness result      | Passed or Review Required                 |
| Face result          | Verified, Review Required or Mismatch     |
| Identity consistency | Consistent or Inconsistent                |
| Risk score           | Value from 0 to 100                       |
| Status               | Verified, Review Required or Mismatch     |
| Explanation          | Text summarising the contributing signals |

#### 4.3.2 Design Considerations

- **Sensitive data:** Extracted personal fields should be stored only where necessary and protected appropriately.
- **Indexing:** Date and status columns should be indexed to keep history filtering fast.
- **Retention:** A retention period should be defined so records are not kept longer than needed.
- **Portability:** Using Django models allows moving from SQLite to PostgreSQL with little change.

### 4.4 User Interface and Result Page

The user interface is designed to be simple so that a reviewer can understand the outcome quickly.

#### 4.4.1 Main Screens

- **Upload screen:** Select a document, check accepted formats and start verification.
- **Live capture screen:** Capture a live photograph for liveness and face checks where applicable.
- **Result screen:** View the verification signals, risk score, status and explanation.
- **History screen:** Browse and filter earlier screenings.

#### 4.4.2 Result Page

The result page presents multiple verification signals. This approach allows the reviewer to understand which part of the verification requires attention. An example result is shown below.

| Check              | Result          |
|--------------------|-----------------|
| Document type      | Aadhaar ✓       |
| OCR                | Review Required |
| QR                 | Unavailable     |
| Document integrity | Medium Risk     |
| Document photo     | Detected ✓      |
| Liveness           | Passed ✓        |

| Check                | Result          |
|----------------------|-----------------|
| Face verification    | Review Required |
| Identity consistency | Inconsistent    |

#### 4.4.3 Usability Guidelines

- Use plain language such as Review Required rather than technical codes.
- Highlight the checks that need attention.
- Show unavailable checks neutrally so they are not mistaken for failures.
- Keep the layout responsive for common screen sizes.

### 4.5 History Management

The History module maintains previous screening records. Users can filter records based on a date range, and the system displays only the records belonging to the selected period.

#### 4.5.1 Information Shown

- Document ID
- Document type
- Date
- Result
- Risk score
- Verification status

#### 4.5.2 Date-Range Filtering

A reviewer chooses a start date and an end date, and the page then lists the screenings within that period along with summary counts. For example:

| Item                     | Example    |
|--------------------------|------------|
| From date                | 01-09-2026 |
| To date                  | 16-09-2026 |
| Total documents screened | 24         |
| Verified                 | 10         |
| Review Required          | 9          |
| Mismatch                 | 5          |

#### 4.5.3 Benefits

- Supports auditing and follow-up of earlier decisions.
- Shows workload and outcome trends over a chosen period.
- Helps tune thresholds by comparing system results with reviewer outcomes.

## 5. Security, Testing and Use Cases

### 5.1 Security, Privacy and Ethical Considerations

Identity documents contain personal and sensitive information, so a screening system must treat security and privacy as core requirements rather than optional extras.

#### 5.1.1 Data Protection

- **Secure transport:** Documents and results should travel only over encrypted connections.
- **Secure storage:** Stored files and records should be protected with access control and, where suitable, encryption.
- **Data minimisation:** Only the information needed for screening should be retained.
- **Masking:** Sensitive identifiers should be masked when displayed wherever the full value is not needed.
- **Retention and deletion:** Records should be deleted when the retention period ends.

#### 5.1.2 Access Control

Only authorised users should be able to upload documents, view results and open the history. Actions should be logged so that access can be audited.

#### 5.1.3 Legal and Regulatory Awareness

Organisations deploying the system should ensure that their use complies with the laws and regulations that apply to them, including data protection law such as India's Digital Personal Data Protection Act, 2023, and any rules that govern the handling of specific identity documents. Users should be informed about why their document is being processed and should give appropriate consent.

#### 5.1.4 Fairness and Ethics

- Results are screening evidence, and no person should be denied a service solely on an automated result.
- Review Required and Mismatch outcomes should be handled respectfully, since they do not prove wrongdoing.
- Performance across different lighting, camera quality and demographic groups should be monitored to detect bias.

### 5.2 Testing Strategy

Testing ensures that each component works correctly on its own and that the complete workflow behaves as intended.

#### 5.2.1 Levels of Testing

- **Unit testing:** Individual functions such as file validation, field parsing and score calculation.

- **Integration testing:** Interaction between the backend, OCR, analysis services and database.
- **System testing:** The full flow from upload to saved result.
- **Usability testing:** Whether reviewers can understand and act on the results.

### 5.2.2 Sample Test Cases

| ID   | Scenario                           | Expected result                                              |
|------|------------------------------------|--------------------------------------------------------------|
| TC1  | Upload a supported PDF or image    | File accepted and processing starts                          |
| TC2  | Upload an unsupported file type    | File rejected with a clear message                           |
| TC3  | Upload a file above the size limit | File rejected with a size message                            |
| TC4  | Upload a clear Aadhaar image       | Type detected as Aadhaar; OCR fields extracted               |
| TC5  | Upload a blurred image             | Confidence lowered; Review Required                          |
| TC6  | Document with unreadable QR        | QR shown as Unavailable, not as fraud                        |
| TC7  | Live capture passes liveness       | Liveness shown as Passed                                     |
| TC8  | Document and live faces differ     | Face verification shows Mismatch or Review Required          |
| TC9  | Signals conflict                   | Identity consistency shown as Inconsistent                   |
| TC10 | Filter history by date range       | Only records within the range are listed with correct counts |

### 5.2.3 Evaluation

Beyond functional testing, system quality should be measured on a representative set of documents, looking at OCR accuracy, correct routing to each outcome, the rate at which documents needing review are caught, and the rate of unnecessary reviews. Reviewer feedback should be used to refine thresholds.

## 5.3 Use Cases and User Roles

This chapter describes who uses VerifyX AI and the typical situations in which they use it. Understanding roles and use cases helps ensure that the interface, permissions and results are designed around real working needs.

### 5.3.1 User Roles

| Role                  | Responsibilities                                                      | Typical actions                                                             |
|-----------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Applicant / Submitter | Provides the identity document and, where required, a live photograph | Upload document, allow camera capture, view submission status               |
| Reviewer              | Examines documents that need a human decision                         | Open results, inspect signals, decide on Review Required and Mismatch cases |

| Role          | Responsibilities                 | Typical actions                                                |
|---------------|----------------------------------|----------------------------------------------------------------|
| Administrator | Manages configuration and access | Set upload limits, manage users, review history and thresholds |

### 5.3.2 Primary Use Cases

| ID  | Use case                                    | Outcome                                                |
|-----|---------------------------------------------|--------------------------------------------------------|
| UC1 | Screen a single identity document           | A structured result with score, status and explanation |
| UC2 | Screen a document with a live photograph    | Liveness and face verification added to the result     |
| UC3 | Review a document flagged for manual review | Reviewer sees which signals need attention and decides |
| UC4 | Browse earlier screenings                   | Filtered history with counts for the selected period   |
| UC5 | Adjust configuration                        | Updated upload limits, formats or scoring thresholds   |

### 5.3.3 Typical Organisational Settings

- **Banks and financial services:** Preliminary screening during customer onboarding before a final know-your-customer decision.
- **Educational institutions:** Checking identity documents submitted during admissions or examinations.
- **Companies:** Screening documents submitted by new employees or vendors.
- **Government and service providers:** Triage of high volumes of submitted documents so staff focus on unclear cases.

### 5.3.4 Reviewer Experience

A reviewer typically opens the history page, filters to the current day or week, and sorts by status so that Mismatch and Review Required records appear first. Opening a record shows the document type, each verification signal, the risk score and the explanation. Because unavailable checks are displayed neutrally, the reviewer can immediately tell the difference between a check that failed and a check that simply could not be performed. After examining the evidence, the reviewer makes a decision and, where the organisation requires it, records the outcome so that thresholds can later be tuned.

For the applicant, the experience is deliberately simple: select a file, follow prompts for a live capture when needed, and receive clear feedback such as a request to upload a clearer image. The system avoids language that suggests wrongdoing, because a Review Required result often reflects image quality rather than any problem with the document.

## 6. Implementation, Deployment and Evaluation

### 6.1 Implementation Details

This chapter describes how the system can be organised in code. The structure below is a proposed organisation that follows the architecture described in Chapter 3; teams should map it to their actual project layout.

#### 6.1.1 Proposed Django Organisation

| Component                  | Responsibility                                                                |
|----------------------------|-------------------------------------------------------------------------------|
| Upload handling            | Receives files, validates type and size, stores them safely                   |
| Classification             | Determines the document type and selects the rule set                         |
| OCR processing             | Runs image pre-processing and text extraction, returns fields with confidence |
| Document analysis          | Evaluates visual characteristics and returns an integrity indication          |
| Verification services      | QR analysis, liveness, face comparison and identity consistency               |
| Risk engine                | Combines signals into a score, status and explanation                         |
| History                    | Stores screening records and serves filtered lists                            |
| Templates and static files | HTML, CSS and JavaScript for the user interface                               |

#### 6.1.2 Example Interface Endpoints

The endpoints below illustrate how the frontend can communicate with the backend. Names are examples only.

| Purpose             | Method | Description                                              |
|---------------------|--------|----------------------------------------------------------|
| Submit document     | POST   | Upload a file and start screening                        |
| Submit live capture | POST   | Send a live photograph for liveness and face checks      |
| Get result          | GET    | Retrieve the result of a screening by document ID        |
| List history        | GET    | Return records, optionally filtered by from and to dates |

#### 6.1.3 File Validation

Before any analysis, the backend checks that the file extension and content type are among the supported formats (PDF, JPG, JPEG, PNG and WEBP) and that the size does not exceed the configured limit. Files that fail these checks are rejected with a clear message. Uploaded files should be stored using generated names rather than user-supplied names, and should never be executed or served in a way that could run embedded content.

### 6.1.4 Image Pre-processing

Before OCR, images can be converted to a consistent size, corrected for rotation, denoised and adjusted for contrast. PDF pages are converted to images so that they follow the same pipeline. Pre-processing improves the quality of extracted text and gives the analysis components more consistent input.

### 6.1.5 Returning Structured Results

Each service returns a small, structured result containing an outcome value, an optional confidence and a short note. The risk engine consumes these structured results, which keeps services independent: an improved OCR component can replace an earlier one without any change to the rest of the pipeline, as long as it returns the same structure.

### 6.1.6 Error Handling

- If a service fails unexpectedly, the corresponding check is marked unavailable and the reason is logged.
- The screening continues with the remaining signals rather than failing completely.
- The final explanation records which checks did not run so the reviewer is informed.

## 6.2 Data Flow and Processing Details

This chapter follows the data through the system, showing what each stage receives, what it produces and what is stored.

### 6.2.1 Stage-by-Stage Data Flow

| Stage                | Receives                                | Produces                                  |
|----------------------|-----------------------------------------|-------------------------------------------|
| Upload               | Original file from the user             | Validated file and a new screening record |
| Classification       | Document image                          | Document type                             |
| OCR                  | Pre-processed image and document type   | Extracted fields with confidence          |
| Document analysis    | Document image                          | Integrity indication                      |
| QR check             | QR region, extracted fields             | Consistency result or Unavailable         |
| Liveness             | Live capture                            | Passed or Review Required                 |
| Face verification    | Document photograph and live photograph | Verified, Review Required or Mismatch     |
| Identity consistency | All available results                   | Consistent or Inconsistent                |
| Risk assessment      | All available results                   | Score, status and explanation             |
| History              | Final result                            | Stored record available for filtering     |

### 6.2.2 Context Level View

At the highest level, the system has three external interactions. The submitter sends a document and, where needed, a live capture. The system returns a status and explanation to the reviewer. The

administrator supplies configuration such as file limits and thresholds. All persistent data is kept in the database and nowhere else.

### 6.2.3 What Is Stored and What Is Not

- **Stored:** The screening record with its results, score, status, explanation and timestamps.
- **Stored with care:** Extracted personal fields and uploaded files, only where needed and with protection.
- **Not needed:** Intermediate images produced during pre-processing, which can be discarded after use.

### 6.2.4 Consistency Between Signals

Identity consistency is the point where separate signals meet. The name, date of birth and document number obtained from OCR are compared with QR content where available, and the face in the document photograph is compared with the live capture. If these agree, confidence rises. If they disagree, the result is marked Inconsistent and the explanation states which signals conflicted. Importantly, a disagreement between two noisy signals may itself be caused by poor image quality, which is why such cases are sent to a reviewer instead of being treated as conclusive.

### 6.2.5 Confidence Handling

Many components can report how sure they are. Low-confidence outputs should not be presented as facts. Instead, they contribute a smaller weight to the risk score and can push the result towards Review Required. This behaviour keeps the system cautious and matches its purpose as a screening tool.

## 6.3 Deployment and Performance Considerations

### 6.3.1 Deployment Options

| Environment       | Database   | Notes                                                  |
|-------------------|------------|--------------------------------------------------------|
| Local development | SQLite     | Simple setup for building and demonstrating the system |
| Testing / staging | PostgreSQL | Closer to production behaviour for realistic testing   |
| Production        | PostgreSQL | Multi-user access, backups and monitoring              |

### 6.3.2 Performance

OCR and image analysis are the most demanding steps. To keep the interface responsive, heavy processing can run in a background task queue while the frontend shows progress and fetches the result when ready. Reducing image size before analysis and reusing loaded models between requests also reduce the time each screening takes.

### 6.3.3 Scalability

- **Stateless web layer:** Multiple copies of the Django application can run behind a load balancer.
- **Separate workers:** Processing services can scale independently of the web layer.

- **Database indexing:** Indexes on date and status keep history queries fast as records grow.

#### 6.3.4 Operations and Maintenance

- **Logging:** Record errors, timings and unavailable checks without storing sensitive content in logs.
- **Backups:** Regular database backups with tested restoration.
- **Model updates:** Version machine-learning components and record which version produced each result.
- **Monitoring:** Track processing time, failure rates and the share of documents in each outcome.

#### 6.3.5 Continuous Improvement

Reviewer decisions provide valuable feedback. When reviewers frequently override a particular outcome, that is a sign that a threshold or component needs attention. Periodic review of these cases, together with new examples of poor-quality images, helps the system improve steadily without changing its basic design.

### 6.4 Illustrative Scenarios

The following scenarios show how the system behaves in typical situations. They are examples to illustrate the workflow, not recorded results.

#### 6.4.1 Scenario A: Clear Document, Live Photo Matches

An applicant uploads a clear image of an identity document and completes a live capture. The document type is detected, OCR extracts the fields with good confidence, the integrity analysis reports Low Risk, the QR code is consistent with the extracted fields, liveness passes and face verification reports Verified. The risk score is low, identity consistency is Consistent and the outcome is Verified. The record is saved and the document can be accepted automatically.

#### 6.4.2 Scenario B: Blurred Image, Unreadable QR

An applicant uploads a photograph taken in poor light. OCR extraction has low confidence and the QR code cannot be read. The QR result is shown as Unavailable, not as a failure. The integrity analysis reports Medium Risk because image quality limits the analysis. The overall outcome is Review Required with an explanation that image quality reduced confidence. The applicant is advised to upload a clearer image, and the reviewer sees exactly why the result was not automatic.

#### 6.4.3 Scenario C: Genuine Document, Different Person

A document appears consistent, but the live capture shows a person whose face does not match the document photograph. Liveness passes, yet face verification reports Mismatch and identity consistency is Inconsistent. The overall outcome is Mismatch / Inconsistent. This scenario shows why a genuine-looking document alone is not sufficient: the multi-signal approach catches a case where a valid document is presented by someone else.

#### 6.4.4 Scenario D: Suspicious Region Detected

The integrity analysis highlights an area of the document that appears inconsistent with the rest of the layout. Other signals are neutral. The result is Review Required with Medium Risk, and the explanation

points the reviewer to the relevant signal. The reviewer can inspect the area directly and decide. The system does not state that the document is forged, because a risk indication is not proof.

#### 6.4.5 Scenario Summary

| Scenario | Key signals                                     | Outcome                 |
|----------|-------------------------------------------------|-------------------------|
| A        | All signals positive                            | Verified                |
| B        | Low OCR confidence, QR Unavailable, Medium Risk | Review Required         |
| C        | Face Mismatch, identity Inconsistent            | Mismatch / Inconsistent |
| D        | Suspicious region, other signals neutral        | Review Required         |

### 6.5 Advantages and Limitations

#### 6.5.1 Advantages

- Reduces manual effort in preliminary screening.
- Provides a centralised verification workflow.
- Supports multiple verification signals.
- Provides structured results.
- Maintains verification history.
- Makes results easier to review.
- Can be extended to additional document types.
- Can support PDF and image-based documents.

#### 6.5.2 Limitations

- Automated screening cannot guarantee the authenticity of every document.
- OCR accuracy depends on document quality.
- Poor lighting, blur, rotation or damaged documents can affect analysis.
- Some verification methods may not be available for every document.
- Official government verification is outside the scope of the system.
- AI/ML predictions should be treated as screening evidence rather than conclusive proof.

#### 6.5.3 Challenges and Mitigation

| Challenge                    | Mitigation                                                     |
|------------------------------|----------------------------------------------------------------|
| Variation in image quality   | Image pre-processing and lower confidence when quality is poor |
| Many document layouts        | Document classification and per-type rules                     |
| Missing or unreadable QR     | Treat as Unavailable, rely on remaining signals                |
| Over-trusting the score      | Show explanations and keep humans in the loop                  |
| Privacy of personal data     | Data minimisation, access control and retention limits         |
| Growth to new document types | Modular rules and services                                     |

## 7. Future Scope, Conclusion and Appendices

### 7.1 Future Scope

The project can be extended in many directions.

- More Indian identity documents.
- International passports and identity documents.
- Multilingual OCR.
- Better document classification models.
- Improved tamper detection.
- More digital and PDF document analysis.
- Additional document-specific verification rules.
- Improved explainable AI.
- Secure institutional verification APIs where legally and technically available.

#### 7.1.1 Suggested Roadmap

| Phase   | Focus                                                                    |
|---------|--------------------------------------------------------------------------|
| Phase 1 | Stabilise Aadhaar and PAN screening; improve OCR accuracy                |
| Phase 2 | Add passports and driving licences with document-specific rules          |
| Phase 3 | Multilingual OCR and improved tamper detection                           |
| Phase 4 | Institutional verification APIs where permitted; stronger explainability |

### 7.2 Conclusion

VerifyX AI provides an AI-assisted approach for preliminary identity document screening. It combines document processing, OCR, verification checks, identity-related checks, risk assessment and verification history into a single platform.

The system is designed to help reviewers identify documents that can be processed automatically and documents that require additional inspection. By presenting evidence clearly, being honest about what could not be checked, and keeping humans responsible for final decisions, it aims to make document screening faster, more consistent and more transparent.

**VerifyX AI — Verify the Document. Verify the Identity**

### 7.3 Glossary

| Term | Meaning                 |
|------|-------------------------|
| AI   | Artificial Intelligence |
| ML   | Machine Learning        |

| Term              | Meaning                                                                               |
|-------------------|---------------------------------------------------------------------------------------|
| OCR               | Optical Character Recognition: converting text in images into machine-readable text   |
| RAG               | Retrieval-Augmented Generation: supplying retrieved knowledge to the explanation step |
| Liveness          | A check that a live capture comes from a real, present person                         |
| Face verification | Comparison of the document photograph with the live photograph                        |
| QR                | Quick Response code                                                                   |
| Risk score        | A value that helps prioritise review                                                  |
| Review Required   | Insufficient confidence for an automatic decision; not proof of a fake document       |
| Unavailable       | A check that could not be performed; not proof of fraud                               |
| Django            | A Python web framework used for the backend                                           |
| OpenCV            | A library for image processing                                                        |

### 7.3.1 Sample Verification Report

| Field                | Value                                                 |
|----------------------|-------------------------------------------------------|
| Document ID          | Example: VX-000123                                    |
| Document type        | Aadhaar                                               |
| Date                 | 16-09-2026                                            |
| OCR                  | Review Required                                       |
| QR                   | Unavailable                                           |
| Document integrity   | Medium Risk                                           |
| Liveness             | Passed                                                |
| Face verification    | Review Required                                       |
| Identity consistency | Inconsistent                                          |
| Risk score           | 64 / 100                                              |
| Status               | Manual Review                                         |
| Reason               | Some verification signals require further inspection. |

### 7.3.2 Technology References

- Python programming language, official documentation.
- Django web framework, official documentation.
- OpenCV library, official documentation.
- SQLite and PostgreSQL, official documentation.
- India Digital Personal Data Protection Act, 2023.